



Transient pores account for cell-penetrating peptide and homeoprotein translocation

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Edited by Makoto Oba, Kyoto Furitsu Ika Daigaku, Kyoto, Japan; received January 23, 2026; accepted May 18, 2026 by Editorial Board Member Francisco Bezanilla

Homeoproteins (HPs) and cell-penetrating peptides (CPPs) enter cells by endocytosis and direct membrane crossing (translocation). However, unlike endocytosis, translocation remains globally unknown. Here, we developed an electrophysiological approach to assess the internalization of CPPs (Tat, R₉, penetratin and R₆W₃) and the HPs Otx2 and En2 through single-cell unitary transient currents in mammalian cells. At resting membrane potential, CPPs or HPs lead to submillisecond transient pores, faster than any endocytosis event, which reveal the rapid passage of the peptide across the membrane i.e., by translocation. We evidenced that expression of specific membrane glycosaminoglycans is mandatory for translocation-induced transient pores. Associated transient currents are supralinearly enhanced by hyperpolarization and poorly affected by depolarization. Moreover, a CPP-conjugated bioactive cargo similarly translocates into cytosol. Finally, we show similar HPs-evoked transient pores in brain cortical pyramidal cells, showing the physiological relevance of translocation, with crucial biotechnological and therapeutic consequences for cell delivery purposes.

cell-penetrating peptide | homeoprotein | translocation | transient pores | membrane potential

Cell-penetrating peptides (CPPs) are short (less than 30 amino acids) and nontoxic molecules, endowed with the ability to cross cell membranes and transport various bioactive cargos aiming to deliver therapeutical diverse molecules into cells (1). Most commonly used CPPs are positively charged and amphipathic peptides, such as penetratin that corresponds to the third helix of the Antennapedia homeoprotein (2) and its short analogue R₆W₃ (3), or positively charged peptides such as Tat derived from the HIV-1 trans-activator of transcription protein (4) or the designed R₉ (5, 6). Homeoproteins (HPs) such as Antennapedia, Otx2, or Engrailed-2 (En2), are also positively charged (basic pI), and known to travel from one cell to the other by an unconventional nonendocytic pathway that remains to be fully elucidated (7). Physiologically, HPs play an essential role during embryogenic development as transcription factors, during the brain critical period of plasticity, and in various pathologies (7). Since CPP applications aim at conveying bioactive cargos inside cells, harnessing their internalization properties is of major interest for biotechnological or therapeutic purposes (8).

The interaction between these positively charged CPPs and HPs and the negatively charged extracellular matrix (essentially made up of proteins, glycoproteins, sulfated proteoglycans and glycosaminoglycans) represents the crucial and initial step of their internalization (9–16). The modalities of CPP internalization remain however strongly debated (17). While endocytosis has been demonstrated, CPPs and HPs can also internalize in conditions precluding endocytosis process through translocation (18), defined as the direct crossing of the membrane bilayer. Namely, CPPs and HPs cellular entry is observed at 4 °C, a temperature at which endocytosis is prevented (10, 19). While endocytosis processes, including clathrin-dependent and -independent pathways or macropinocytosis, take place in a temporal range spanning from tens of seconds to subsecond scale in nonneuronal cells (20), translocation is hardly tracked or quantified separately from endocytosis in living cells. Although dedicated biotechnological and engineering methods have been developed to evidence translocation of HPs and CPPs (21–23), the existence of this cell entry pathway remains to be solved (18). In addition, it has been suggested that oligoarginine CPP translocation could occur at a submicrosecond time-scale, as suggested by a debated (24) molecular dynamics modeling study (25, 26), making its observation more challenging.

More than two decades ago, the membrane potential was reported as a key parameter for the cell entry of CPPs (27). Various CPPs show decreased internalization in depolarized cells, whereas hyperpolarization fostered their entry (28, 29). In addition, the CPP Pep-1

Significance

We used a single-cell double patch-clamp method to study the direct translocation of cell-penetrating peptides (CPP) and homeoproteins (HP) across mammalian cell membranes. This approach, applicable to any cell type, revealed similar translocation of HPs and CPPs through submillisecond transient pores—far faster than endocytosis pathways—with similar kinetics in cell lines and brain pyramidal cells. Hyperpolarization dramatically enhanced translocation (10 to 100×), while depolarization had minimal effect. Translocation was confirmed by fluorescence imaging at 22 °C, delivery of a toxic cargo (KRKLAK) into cells and electrophysiological recordings at 11 °C. Pore formation required cell-surface glycosaminoglycans, which interact with peptides to facilitate membrane crossing. These findings clarify the biophysical mechanism of CPP/HP entry, offering promising avenues for targeted drug delivery and biotechnological applications.

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This article contains supporting information online at <https://www.pnas.org/lookup/suppl/doi:10.1073/pnas.2602649123/-/DCSupplemental>.

Published June 22, 2026.

induces ionic fluxes in *Xenopus* oocytes and hyperpolarized model lipid membranes (30).

Here, we sought to detect the occurrence of internalization events and characterize their properties. For this purpose, we set up single-cell double patch-clamp recording, allowing submillisecond temporal resolution with one CPP/HP-containing pipette in cell-attached mode and the second pipette dedicated to whole-cell configuration to control membrane potential. It enables to track fast unitary events while varying CPPs or HPs concentration and membrane potential. Altogether, we show unitary translocation pores induced by CPPs and HPs, and characterized by a higher frequency and faster duration than any endocytosis process.

Results

Detection of CPP-Induced Currents With Single-Cell Double Patch-Clamp. We aimed at characterizing the internalization of CPPs (Tat, R₉, penetratin, and R₆W₃), HPs (En2 and Otx2), and a poorly internalized peptide R₆L₃, at a submillisecond time-scale and single-cell level. To do so, we set up an electrophysiological approach (Fig. 1 A and B) to study the internalization of those CPPs and HPs by double patch-clamp recordings at a single-cell level in CHO-K1 cells and derivatives, and in cortical pyramidal neurons in acute mouse brain slices. The double patch-clamp consisted in one patch-clamp pipette dedicated to whole-cell recordings in the voltage-clamp mode to control the holding membrane potential (E_{holding}), and a second patch-clamp pipette in the cell-attached configuration containing the CPP or HP (at various concentrations) to monitor the ion fluxes induced by internalization current transients (Fig. 1 A and B and *Materials and Methods*). The cell-attached pipette contained the CPP or HP diluted in cerebrospinal fluid-like solution and contacted about $1\ \mu\text{m}^2$ of the plasma membrane. This limited surface of membrane under the cell-attached favored the detection of unitary events. Indeed, cell-attached recordings have been widely used for characterization of single channel currents and are particularly well-suited for the detection of brief current transients without affecting the cell physiology. The giga-Ohm seal-isolated patch of membrane achieved for cell-attached recordings allowed us to detect any submillisecond event (sampling rate of 20 kHz) in the range of few pA occurring locally under the cell-attached pipette. The recording cell-attached pipette was held at a potential corresponding to a null holding current in order to avoid bias in the currents recorded via this electrode (31).

As the main cellular model, we chose the ovarian cells CHO-K1 widely used to study CPP internalization (32). Moreover, CHO-K1-derived mutant cell line with about 90% reduced expression of heparan (HS) and mono-sulfated chondroitin sulfates (CS) GAGs (pgsA-735) (33), and a cancer cell line (SKOV3) that, on the opposite, overexpresses (4,6)-disulfated chondroitin GAGs (34), offer an advantage of allowing the analysis of the role of GAGs in the same cell type. Finally, as a physiological integrated model, we also tested HP internalization in acute brain slices in pyramidal cells of the somatosensory cortex of mice. All recordings were performed at 34 °C (except *SI Appendix, Fig. S4*).

Whole-cell recordings showed that CHO-K1, in the absence of CPPs or HPs, displayed a linear current-voltage (I-V) relationship with an input resistance (R_i) at $362 \pm 56\ \text{M}\Omega$ and a resting membrane potential (RMP) at $-26 \pm 5\ \text{mV}$ ($n = 64$ cells) (Fig. 1B). In the cell-attached side, Fig. 1B shows an example of cell-attached recording, with a raw current trace resulting from the presence of R₉ (10 μM) in the pipette and the analysis process to extract CPP- and HP-induced current. We first calculated local current trend

(green line in Fig. 1B) by computing the 500 ms-width moving average of each current data point. We then computed the average of the SD of all the recordings performed at a reference E_{holding} ($-25\ \text{mV}$) in the same cell. Finally, each time point of the recording larger than the sum of the local trend at this time point and the average SD at the reference E_{holding} (red line in Fig. 1B) was identified as a transient current event. Events formed by less than 10 consecutive event points (i.e., $<500\ \mu\text{s}$) were discarded and considered electrical background noise (*Materials and Methods*).

We evaluated the internalization of Tat, R₉, penetratin and R₆W₃ together with the poorly cell-penetrating R₆L₃ peptide (35). Tat and Penetratin were chosen as protein-transduction domains derived respectively from the HIV-Tat protein and the *Antennapedia* homeoprotein. R₉ and R₆W₃ were taken as Tat and penetratin short analogues, respectively. Tat and R₉ peptides are essentially arginine-rich peptides whereas penetratin and R₆W₃ are amphipathic peptides. Interestingly, the charge density (number of positive charges/numbers of amino acids) of these peptides was different, being 1 for R₉ (RRRRRRRRR), 0.80 for Tat (GRKKRRQRRR), 0.67 for R₆W₃ (RRWRRRWR) or R₆L₃ (RLLRLRLR) and 0.38 for penetratin (RQIKIWFQNRRMKWKK). Regarding HPs, we chose Engrailed-2 (En2) (Uniprot. Q05917, 289 AA, pI = 9.95, net charge +13) and Otx2 (Uniprot. Q8JG31, 289 AA, pI = 9.41, net charge +10), two HPs used to study and evaluate HP internalization (36).

Nano- to Micromolar CPP Concentrations Induce Fast Transient Currents at RMP. We first investigated CPP internalization at $E_{\text{holding}} = -25\ \text{mV}$ i.e., close to the CHO-K1 RMP (Fig. 1B) and monitored the induced currents by varying CPP concentration; a representative raw current trace recorded via the cell-attached pipette containing R₆W₃ (10 μM) is illustrated in Fig. 1C. All peptides, except R₆L₃, promote detectable ion fluxes (Fig. 1D–H) as indicated by the significant changes in the number of membrane events (*SI Appendix, Fig. S1*) and in the total charge transfer that corresponds to the integral of currents recorded during the 2-s of $E_{\text{holding}} = -25\ \text{mV}$ (Fig. 1D–H). Overall, the total charge transfer and number of events increased upon increasing CPP concentration. The synthetic peptides R₉ and R₆W₃ at 10 μM induced globally more events and larger total charge transfer than their respective natural protein-transduction domains, Tat and penetratin. Noticeably, when compared to the null CPP condition, we measured a significant total charge transfer for concentrations as low as 1 μM for Tat and penetratin and 100 nM for R₆W₃ and R₉. Regarding R₆W₃, we could even detect unitary events at 10 nM. We reached the limit of event detection for the lowest R₆W₃ concentration tested, i.e., 1 nM, since the total charge transfer and number of events measured were not significantly different from background noise without peptide (Fig. 1H and *SI Appendix, Fig. S1*). Interestingly, R₆L₃ at various similar concentrations (from 100 nM to 10 μM) did not generate such detectable events, in line with its poor cell-penetrating properties (35). The R₆L₃ experiment shows the robustness of our electrophysiological approach allowing to distinguish transient events due to peptide internalization from background noise.

We further analyzed the unitary transient currents for concentrations in which unitary events can be isolated (Fig. 1I), i.e., 1 μM for Tat, penetratin and R₆W₃ and 0.1 μM for R₉ (for larger concentrations, unitary events merge altogether to form macroscopic currents). The amplitudes of single events did not depend on the nature of the CPP, i.e., 2.7 ± 0.1 , 2.4 ± 0.03 , 3.1 ± 0.04 , and $2.3 \pm 0.08\ \text{pA}$ for Tat (1 μM), R₉ (0.1 μM), penetratin (1 μM), and R₆W₃ (1 μM), respectively. p_{on} and p_{off} represent a proxy for event frequency and duration, respectively (*Materials and Methods*). We found that the duration of the CPP-induced events was mainly in

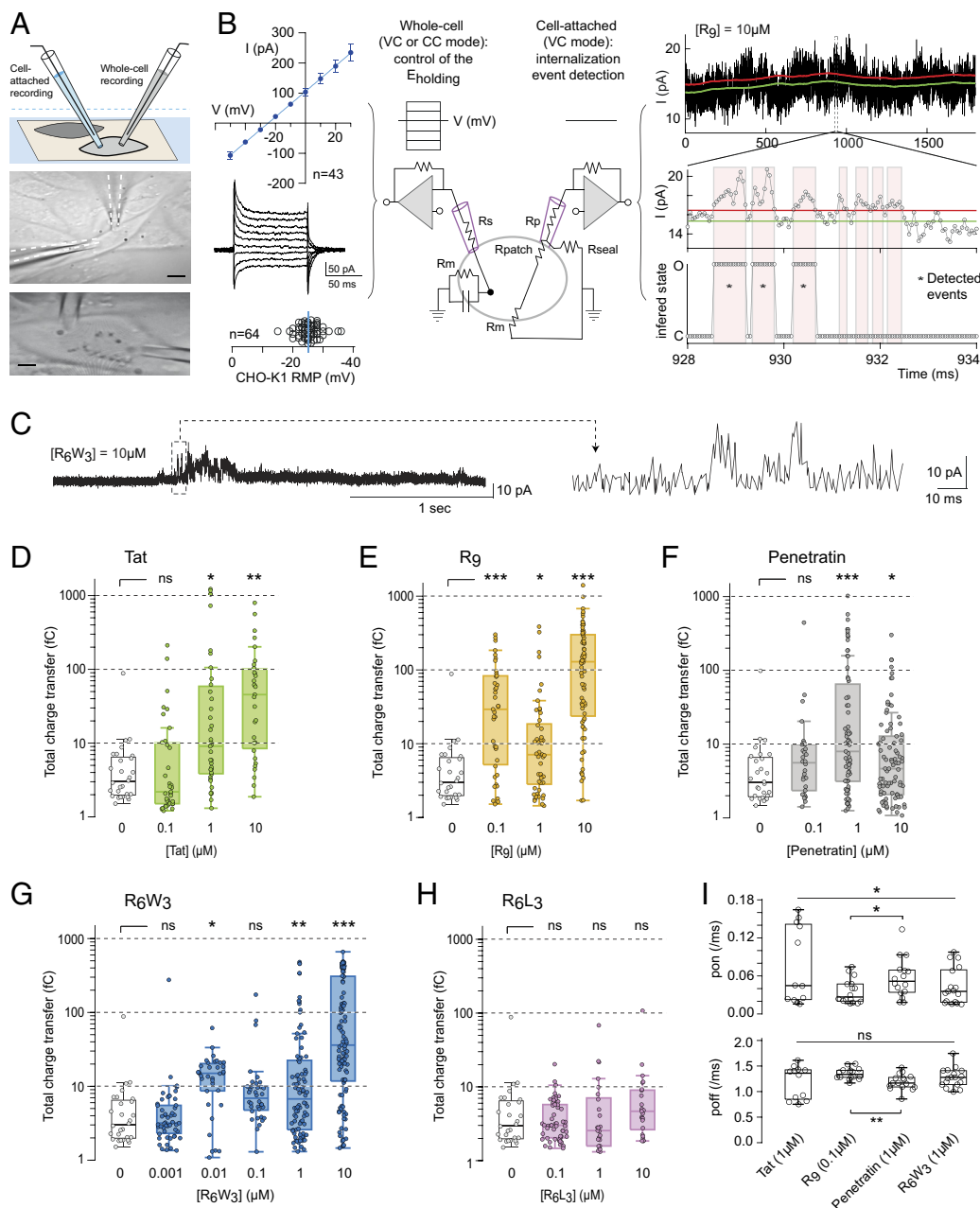


Fig. 1. Micromolar concentrations of CPPs induced transient currents indicative of submillisecond transitory pores. (A) Illustration and infrared microphotographs of double patch-clamp recordings of a single CHO-K1 cells. (Scale bars, 5 and 1 μm .) (B) Equivalent recording circuit model overlaid on a schematic experiment (central panel) with one pipette performing a whole-cell recording (Left) while the CPP-containing pipette records transient currents in cell-attached configuration (Right). Left panels: whole-cell recordings showing a linear I/V relationship, with representative current traces, and RMP -26 ± 5 mV ($n = 64$ cells); Right panels: cell-attached recordings showing single raw current trace (R_9 at 1 μM) with the methodology overview to extract elementary events (Materials and Methods). (C) Raw current traces recorded via the cell-attached electrode showing induced currents due to the internalization of R_6W_3 (10 μM). (D–H) Total charge transfer measured at $E_{\text{holding}} = -25$ mV induced by (D) Tat (0.1 μM : $n = 5$ cells; 1 μM : $n = 4$; 10 μM : $n = 4$; $p_{\text{ANOVA}} = 9.26 \times 10^{-3}$), (E) R_9 (0.1 μM : $n = 5$; 1 μM : $n = 5$; 10 μM : $n = 5$; $p_{\text{ANOVA}} = 3.69 \times 10^{-9}$), (F) penetratin (0.1 μM : $n = 4$; 1 μM : $n = 6$; 10 μM : $n = 5$; $p_{\text{ANOVA}} = 5.84 \times 10^{-4}$), (G) R_6W_3 (0.001 μM : $n = 5$; 0.01 μM : $n = 4$; 0.1 μM : $n = 4$; 1 μM : $n = 6$; 10 μM : $n = 6$; $p_{\text{ANOVA}} = 1.02 \times 10^{-15}$) and (H) R_6L_3 (0.1 μM : $n = 4$; 1 μM : $n = 4$; 10 μM : $n = 5$; $p_{\text{ANOVA}} = 0.307$). The null concentration data were obtained with regular extracellular solution without CPP in the cell-attached pipette ($n = 5$ cells). Statistics indicate Student's t test values comparing the null concentration with the indicated CPP concentration (bilateral unpaired test, using Satterthwaite's approximation if unequal variances from F -test). ns: $P > 0.05$, * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$. Circle-dot: individual sweep. (I) p_{on} and p_{off} of current events as a proxy for event frequency and duration, respectively, were extracted by Bayesian inference (Materials and Methods) for each tested CPP at a minimal concentration (1 or 0.1 μM)

the submillisecond scale. Interestingly, the opening duration of the transient pores was not significantly different among all tested CPP (one-way ANOVA, $P = 0.0873$) (Fig. 1I).

Overall, we found a marked concentration dependency of CPP-induced currents and revealed cell-penetrating activity for concentrations in the nanomolar range, i.e., at concentrations at least 100-times lower than those classically studied. In addition, the kinetics of these transient currents indicate a translocation mechanism of CPP entry since the duration of such CPP-induced transient currents is in a submillisecond range, when within the hundreds of milliseconds time-scale for endocytosis processes (20).

CPP Internalization Assessed With Two-Photon Imaging. We performed two-photon imaging experiments combined with single cell-attached patch-clamp recordings (SI Appendix, Fig. S2) to visualize CPP entry. We chose R_6W_3 tagged with the fluorophore

carboxyfluorescein (CF- R_6W_3) and carboxyfluorescein (CF) alone as the corresponding control. Due to the technical limitations to insert patch-clamp pipette under the two-photon system, we performed cell-attached recordings, without the whole-cell side, which left cells at their RMP. We chose to test the 10 μM concentration for which potent transient currents were recorded for all CPPs. CF- R_6W_3 or CF was placed within the cell-attached pipette and the two-photon imaging recordings started ~ 5 s after establishment of the giga-seal. We could detect fluorescence within the cytosol few tens of seconds after giga-seal establishment with CF- R_6W_3 , whereas the fluorescence of CF alone stayed restricted to the patch-clamp pipette and never crossed the cell membrane (SI Appendix, Fig. S2). With CF- R_6W_3 , the cytoplasmic fluorescence was significantly observed sporadically from 22 s and then consistently from 51 s until the end of the cell-attached recordings (100 s) in contrast to CF alone, and exhibited a linear increase over time.

We ensured that the giga-Ohm seal was fully preserved during the cell-attached configuration experiment (*SI Appendix, Fig. S2*). Indeed, when after 100 s of recordings we switched from the cell-attached to whole-cell configuration, both CF-R₆W₃ and CF invaded rapidly and massively the cell cytosol. In this latter condition, the fluorescence signal reaches equilibrium between the pipette solution and the intracellular milieu after 3 TO 5 s (*SI Appendix, Fig. S2*).

Overall, our results show that CPPs internalized at a micromolar concentration with a cytoplasmic fluorescence detection from 22 s after CPP exposure, and that ion fluxes observed in *Fig. 1* are more likely linked to the CPP entry into the cell.

CPP-Evoked Events Are Boosted By Membrane Hyperpolarization.

Since physiologically the membrane potential is subjected to large range of voltage variations (e.g., neurons display subthreshold variations from -40 to -90 mV) and CPPs internalize in any cell type, we examined whether Tat-, R₉-, penetratin-, and R₆W₃-induced pores depend on the membrane potential. To do so, we imposed hyperpolarized or depolarized voltage steps via the whole-cell patch-clamp electrode, holding the membrane potential at either -75, -55, -40, -25, -10, or +5 mV (*Materials and Methods*). We observed that hyperpolarized steps increased considerably the total charge transfer and CPP-induced currents (*Fig. 2*). This is illustrated by raw CPP-induced currents at $E_{\text{holding}} = -55$ and -75 mV (*Fig. 2 A1–E1*), quantified by the relationship of total charge transfer across various E_{holding} (*Fig. 2 A2–E2*) and the total charge transfer ratios (*Fig. 2 A3–D3*). The increase of the total charge transfer reaches maximal values at the most hyperpolarized potential, i.e., $E_{\text{holding}} = -75$ mV, with different magnitudes depending on the CPP sequence: about 8,000 fC for Tat, 3,000 fC for R₉, 1,200 fC for penetratin and R₆W₃, in contrast to 10 fC recorded for R₆L₃. This stepwise voltage-dependent increase of the CPP-induced total charge transfer was globally related to peptide concentrations (*SI Appendix*). Regarding R₆L₃, we did not observe any voltage-dependency of the sparse CPP evoked-currents, regardless of the peptide concentration, with the total charge transfer remaining below 10 fC (*Fig. 2E2*), which is far below what we recorded with all other peptides.

Interestingly, a longer peptide (R₁₀W₆) displayed identical pore formation properties (*SI Appendix, Fig. S3*) as its shorter analogue R₆W₃ since they share identical p_{on} (0.035 ms⁻¹) and p_{off} (1.1 ms⁻¹) values at RMP; at -75 mV, the value of total charge transfer induced by 10 μM of R₆W₃ or R₁₀W₆ is 1,200 fC in both cases.

We next tested the adverse hypothesis, meaning that these voltage-dependent variations in the total charge transfer values could reflect the ion fluxes driven only by the electrochemical gradient, and not by additional CPP entry and pore formation. To decipher the relative contribution of these interdependent two components, we next determine whether the stepwise potential-dependent modification of the total charge transfer followed a sub- or supralinear relationship with the ion electromotive force, i.e., the difference between the membrane potential and the 0 mV inversion potential of nonspecific pores (see *SI* for the distinction between electromotive force effect and pore opening modification). We calculated the total charge transfer ratio (normalized to the total charge transfer value at -25 mV) (*Fig. 2 A3–D3*) and found that hyperpolarization boosted CPP-induced currents, with a 10- to 100-times increase in internalization. Indeed, at 10 μM penetratin internalized 20 to 30-times more at -75 and -55 mV, respectively, when Tat internalized 100-times more. Even for lower concentration, R₉ at 1 μM displayed 30-, 50-, and 100-times more total charge transfer at -40, -55, and -75 mV, respectively. Interestingly, for 100 nM, a 10-times increase is reached in several hyperpolarized conditions for Tat or R₆W₃. Overall, our results

also suggest that extra pools of pores would be formed upon hyperpolarization. To distinguish the effect of the electrochemical gradient flowing through a constant number of pores induced by CPP from a sub- or supralinear effect of the membrane potential on CPP entry on the numbers and sizes of the pores, we computed theoretical values based of the total charge transfer measured at $E_{\text{holding}} = -25$ mV (*Materials and Methods*) (*Fig. 2 A3–D3* and *SI Appendix, Table S1*). Briefly, the theoretical charge transfer ratio (*Fig. 2*) is the estimated charge transfer ratio if the pore opening (size and number) is not modified compared to RMP, thus only the electromotive force varied. A positive difference with this theoretical ratio indicates that additional pores and/or larger pores are opened at a given potential compared to RMP. By doing so, we can thus distinguish the effect of the sole electromotive force from additional and/or modified pore opening. These theoretical values draw the linear relationship based on a constant number of pores (observed at -25 mV) with increasing electrochemical gradient. We observed a supralinear effect of the membrane potential on the total charge transfer, regardless of the concentration, indicative of additional pores formed by CPPs. This is exemplified by the much higher charge transfer (25-times) for 10 μM Tat at -75 mV, than the one given by the theoretical values. As a control, R₆L₃ did not show supralinear relationship (*SI Appendix, Table S1*).

Overall, our data show that CPP internalization is favored by hyperpolarization, whereas hyperpolarization is known to have no effect on endocytosis (37), further supporting that we detected herein direct CPP translocation across the cell plasma membrane via transient pores. The membrane potential exerts an asymmetrical effect on CPP translocation boosting the internalization by one or two log-factors upon hyperpolarized states (*Fig. 2 A3–D3*).

Duration of CPP-Evoked Unitary Currents Remains Unaffected by Lowering Temperature. To further analyze whether the CPP-induced currents reflect a translocation process, and not endocytosis, we compared CPP-induced transient currents at 34 °C and 11 °C. All endocytic processes are abolished at temperature lower than 15 °C (38), while translocation is expected to decrease, in terms of number of events because of reduced membrane fluidity (39).

We lowered the temperature down to 11 °C and measured the R₆W₃- and penetratin-evoked currents (*SI Appendix, Fig. S3*). Because of the expected drop in the total charge transfer at 11 °C, we tested R₆W₃ and penetratin at 10 μM. In these conditions, we could still detect CPP-evoked currents, which were reduced (mean total charge transfer reduced ~3-times compared to 0.01 μM R₆W₃ or 0.1 μM penetratin) but not abolished, compared to 34 °C (*SI Appendix, Fig. S3*).

Regarding R₆W₃, the rate of sparse events observed for 10 μM R₆W₃ at 11 °C is reminiscent of the rate of events observed for 10 or 100 nM R₆W₃ at 34 °C temperature (*SI Appendix, Fig. S3A*). Accordingly, we compared statistically those CPP-induced sparse unitary events in these conditions and estimated their kinetic parameters p_{on} and p_{off} . The kinetic parameters $1/p_{\text{on}}$ indicate the mean delay between the occurrence of two unitary events (ms) and $1/p_{\text{off}}$, their duration (ms). For R₆W₃, when p_{on} was reduced about 3-times upon decreasing the temperature from 34 °C (10 and 100 nM) to 11 °C (10 μM), p_{off} remained unchanged. This implies that the occurrence of R₆W₃-induced currents is about one order of magnitude lower at 11 °C (~2 Hz) than at 34 °C (~6 Hz), but with a similar duration (~500 μsec) regardless of the temperature, 11 °C vs 34 °C. At higher concentrations such as 1 μM R₆W₃ at 34 °C, transient evoked-currents were about 10-times more frequent (~100 Hz) and their duration slightly increased up to 800 μsec compared to those recorded for lower CPP concentrations. Similar results were obtained with penetratin, for which events

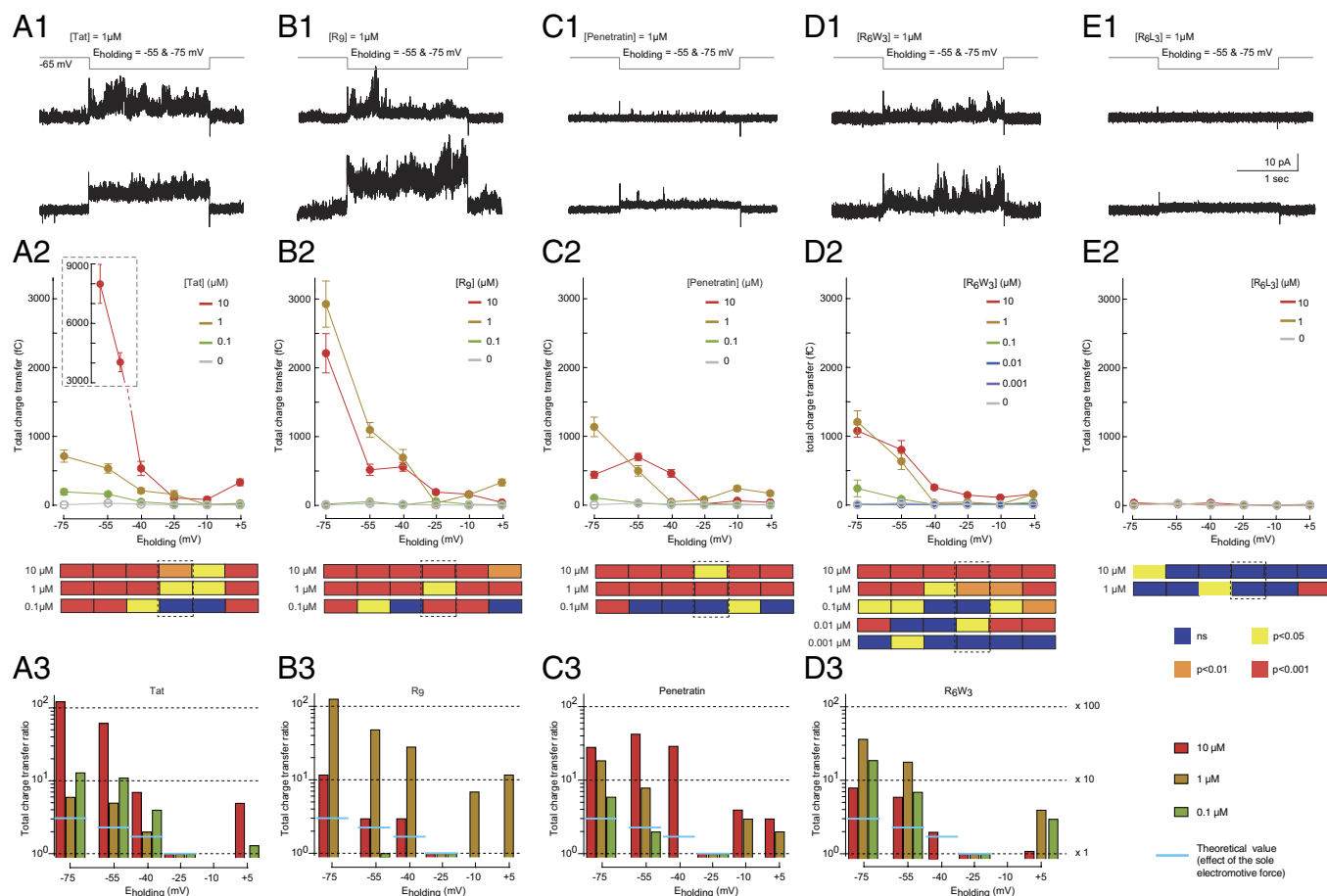


Fig. 2. Membrane hyperpolarization favors CPP-evoked events. (A1–E1) Representative raw current traces recorded in cell-attached configuration show the CPP-induced currents for $E_{\text{holding}} = -55$ and -75 mV, for (A1) Tat, (B1) R₉, (C1) penetratin, (D1) R₆W₃ and (E1) R₆L₃ (1 μM). (A2–E2) Total charge transfer measured at various E_{holding} (-75 , -55 , -40 , -25 , -10 , and $+5$ mV) induced by the indicated concentrations of Tat (0.1 μM : $n = 5$ cells; 1 μM : $n = 5$; 10 μM : $n = 4$), R₉ [0.1 μM : $n = 5$; 1 μM : $n = 5$; 10 μM : $n = 5$; $p_{\text{ANOVA}} = 6.39 \times 10^{-29}$ (concentration effect), 2.02×10^{-56} (E_{holding} effect)], penetratin [0.1 μM : $n = 5$; 1 μM : $n = 4$; 10 μM : $n = 5$; $p_{\text{ANOVA}} = 1.97 \times 10^{-29}$ (concentration effect), 9.23×10^{-42} (E_{holding} effect)], R₆W₃ [0.001 μM : $n = 5$; 0.01 μM : $n = 5$; 0.1 μM : $n = 5$; 1 μM : $n = 6$; 10 μM : $n = 6$; $p_{\text{ANOVA}} = 5.61 \times 10^{-40}$ (concentration effect), 1.35×10^{-50} (E_{holding} effect)] and R₆L₃ [0.1 μM : $n = 5$; 1 μM : $n = 4$; 10 μM : $n = 5$; $p_{\text{ANOVA}} = 0.051$ (concentration effect), 6.44×10^{-12} (E_{holding} effect)]. The null concentration data (“0”) was obtained without CPP in the cell-attached pipette solution ($n = 5$ cells) at each tested E_{holding} : -75 , -55 , -40 , -25 , -10 , and $+5$ mV. Color-coded statistics below each graph indicate Student’s t test values comparing the null concentration with the indicated CPP concentration (bilateral unpaired test, using Satterthwaite’s approximation). (A3–D3) The total charge transfer at a specified E_{holding} normalized to the total charge transfer value recorded at $E_{\text{holding}} = -25$ mV shows the changes of total charge transfer of order of magnitude, and that hyperpolarization favored by a factor 10 or even 100 times the CPP internalization, in particular Tat and R₉. Theoretical values are given in the *SI Appendix, Supplementary Tables*. Blue line on each three-bars block shows the theoretical charge transfer ratio expected for a constant number of pores (determined at -25 mV) with the only effect of the electrochemical gradient (*Materials and Methods*). This indicates that the measured charge transfer ratio follows a supralinear relationship, showing that the peptides induce extra pore formation. For the distinction between electromotive force effect and pore opening modification, see *SI Appendix*.

occurred at 2 Hz and last for ~ 600 μsec both for the 11 $^{\circ}\text{C}$ –10 μM and the 34 $^{\circ}\text{C}$ to 100 nM conditions (*SI Appendix, Fig. S3B*).

Altogether, our data indicate that lowering the temperature allows to unveil the origin of the recorded transient currents, which exclude the involvement of endocytosis processes. By lowering the temperature, we observed a decrease from 10 to 100-times of the occurrence of CPP-induced currents but with an unaffected opening duration, i.e., ~ 500 to 800 μs . The duration of the CPP-evoked single events (~ 500 to 800 μs) is about 1,000-times more rapid than the frequency and time-scale duration of the classical and fast endocytosis (about 2 min and 1 to 60 s, respectively) and even ultra-fast (50 to 300 ms) processes (20, 40, 41). These results indicate that CPPs translocate across the cell membrane by forming similar transient pores.

Cell-surface GAGs are Mandatory for the Occurrence of CPP-Evoked Currents. We have previously determined with quantitative mass spectrometry that cell-surface sulfated GAGs play a key role in CPP internalization (10, 14, 42, 43). Here, we explored the effect of

GAGs in the occurrence of CPP-evoked currents at a submillisecond resolution. To this end, we used two cell lines in which GAGs were mostly deprived (pgsA-745) or overexpressed (SKOV3) (Fig. 3A). First, we evaluated CPP entry into pgsA-745 cells, the engineered CHO cell line deficient in heparan and chondroitin sulfate glycosaminoglycans (Fig. 3B–E). The pgsA-745 cell line expresses 5 to 10% of heparan and chondroitin monosulfate levels compared to the wild-type CHO-K1 cells, a deficiency reported to be detrimental for cell-penetration (32). pgsA-745 cells displayed a RMP = -25 ± 4.5 mV ($n = 41$) with $R_i = 349 \pm 43$ M Ω similar to CHO-K1 cells ($P = 0.637$ and $P = 0.866$ for RMP and R_i , respectively). We conducted voltage-dependent exploration of Tat, R₉, penetratin, and R₆W₃, in pgsA-745 and observed considerably reduced CPP-induced currents (Fig. 3B–E). In addition, those sparse CPP-evoked currents were mostly independent from E_{holding} , even at -75 mV. Our results indicate that cationic (Tat, R₉) and amphipathic (penetratin, R₆W₃) peptides translocation is fully dependent on the presence of cell-surface GAGs. Noticeably, a highly hyperpolarized membrane (-75 mV) could not compensate for the absence of

cell-surface GAGs as we did not observe CPP-evoked currents with this voltage-imposed condition (*SI Appendix, Table S2*).

We then asked whether the opposite situation, namely glycosaminoglycan overexpression at the cell-surface, favors the occurrence of CPP-evoked currents. We chose the SKOV3 cell line which overexpresses chondroitin (4,6)-disulfate and heparan sulfates (34, 44) and tested R_6W_3 (Fig. 3E). SKOV3 cells have a RMP value of -28 ± 7 mV ($n = 39$ cells) with $R_i = 380 \pm 36$ M Ω , similar to CHO-K1 cells ($P = 0.268$ and $P = 0.764$ for RMP and R_i , respectively). The total charge transfer induced by R_6W_3 was enhanced supralinearly up to 4-times for the most hyperpolarized condition (-75 mV) in SKOV3 cells, when compared to CHO-K1 cells (Fig. 1E and *SI Appendix, Table S2*); see also representative

raw current traces for R_6W_3 10 μ M in CHO-K1, pgsA-745, and SKOV3 cell lines (Fig. 3E). For the other membrane potential, we recorded similar total charge transfer values in SKOV3 compared to CHO-K1 cells.

Altogether, our data show that translocation of CPPs strongly depends on the presence of glycosaminoglycans (43) and that hyperpolarized membrane states hardly compensate for the absence of these cell-surface negatively charged polymers.

HPs, En2 and Otx2, Evoked Transient Currents Similar to Those of CPPs, and Dependent on Cell-Surface Glycosaminoglycans. Since penetratin and R_6W_3 are derived from HPs, we evaluated next whether En2 and Otx2 HPs exhibited similar phenotypes as these

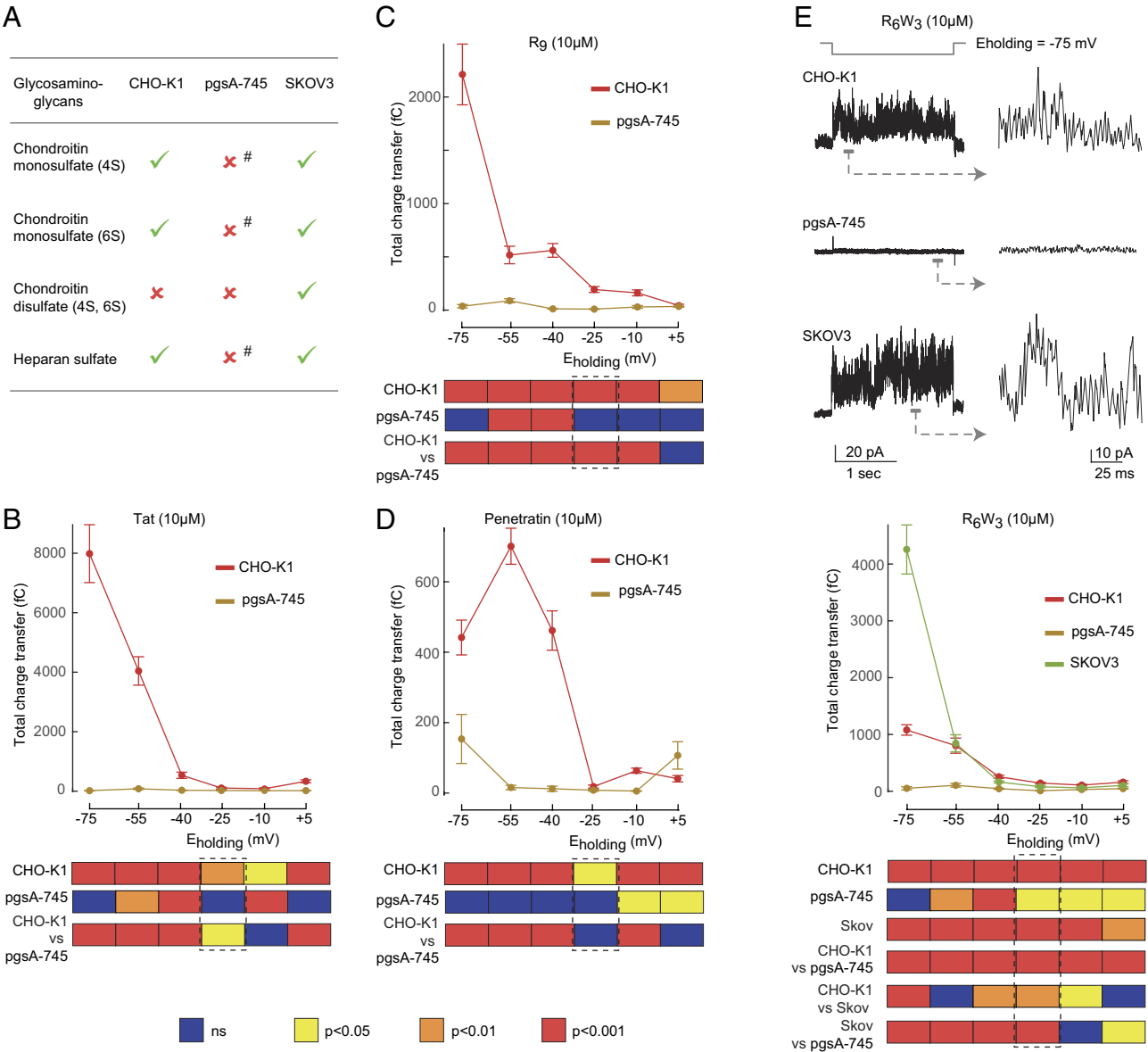


Fig. 3. Cell-surface GAGs are mandatory for the occurrence of CPP-evoked currents. (A) Chondroitin and heparan sulfates expression in CHO-K1, pgsA-745, and SKOV3 cell lines. # Indicates that low levels can be found (5 to 10% compared to CHO-K1). GAGs are expressed at very low levels in pgsA-745 and overexpressed in SKOV3. (B–D) Total charge transfer measured at various E_{holding} induced by CPPs (B) Tat, (C) R_g and (D) penetratin at 10 μ M in pgsA-745 compared to CHO-K1 (Tat: $n = 4$ and 5, $p_{\text{ANOVA}} = 1.34 \times 10^{-23}$ (E_{holding} effect), 2.17×10^{-25} (cell line effect), R_g : $n = 5$ and 4, $p_{\text{ANOVA}} = 2.67 \times 10^{-24}$ (E_{holding} effect), 9.44×10^{-21} (cell line effect), and penetratin: $n = 5$ and 5, $p_{\text{ANOVA}} = 1.08 \times 10^{-22}$ (E_{holding} effect), 4.23×10^{-28} (cell line effect), in CHO-K1 and pgsA-745 cells, respectively). (E) Representative raw current traces recorded in cell-attached configuration show the R_6W_3 -induced currents at 10 μ M for $E_{\text{holding}} = -75$ mV in CHO-K1, pgsA-745, and SKOV3. Total charge transfer measured at various E_{holding} induced by R_6W_3 at 10 μ M in pgsA-745 ($n = 4$) and SKOV3 ($n = 4$), and compared to CHO-K1 ($n = 6$) [$p_{\text{ANOVA}} = 9.81 \times 10^{-59}$ (E_{holding} effect), 2.25×10^{-12} (cell line effect)]. Overall, CPP-induced currents are considerably reduced in pgsA-745 cells whereas increased in SKOV3. The null concentration data used for the statistics were obtained without CPP in the cell-attached pipette solution at each tested E_{holding} in CHO-K1 ($n = 5$), pgsA-745, ($n = 5$) and SKOV3 ($n = 4$) cells. Color-coded statistics above each graph indicate Student's t test values comparing the null concentration with the indicated CPP concentration (bilateral unpaired test, using Satterthwaite's approximation if unequal variances from F -test).

CPPs in terms of concentration-, voltage-, and glycosaminoglycans-dependence, as well as the transient current characteristics (p_{on} and p_{off}).

In CHO-K1 at RMP ($E_{\text{holding}} = -25$ mV), we detected Otx2- and En2-evoked currents for 20 μM HP concentration, but not for lower concentration such as 10 μM (Fig. 4A and B). This result highlights a major concentration shift ($\times 100$ to 1,000), compared to their derived amphipathic peptides, penetratin, and R_6W_3 . Interestingly, En2- and Otx2-induced currents exhibited an asymmetric voltage-dependency (Fig. 4A and B), similarly to the CPPs: hyperpolarizing steps favored the occurrence of HP-induced transient currents, whereas depolarizing steps did not influence the occurrence and magnitude those currents compared to RMP. It should be noted that at -75 mV, we recorded a potent increase of 25- and 45-times for total charge transfer-evoked by En2 (20 μM) and Otx2 (50 μM), respectively. As observed for CPPs, the measured charge transfer ratio is supralinearly higher than the theoretical charge transfer ratio (SI Appendix, Table S3), indicating that HPs also induce the formation of additional transient pores upon hyperpolarization.

We further analyzed the kinetics (p_{on} and p_{off}) of unitary transient currents for En2 and Otx2 (20 μM , $E_{\text{holding}} = -25$ mV) and found amplitudes of single events of 2.8 ± 0.1 and 2.0 ± 0.1 pA, respectively. p_{on} and p_{off} calculated for En2 and Otx2 indicate that unitary events displayed durations and frequencies similar to those found for CPPs, i.e., in the submillisecond range (p_{off}) and a 12 to 16 Hz frequency range (p_{on}) (Insets in Fig. 4A and B, compared to Fig. 1 and SI Appendix, Fig. S2 for CPPs). Notably, the p_{off} for En2 and Otx2 were not significantly different from those of Tat, R_9 , penetratin, R_6W_3 (one-way ANOVA for p_{off} : $P = 0.119$) indicating a similar mechanism of translocation, i.e., transitory pores based on the submillisecond opening duration of single events.

We then tested how the overexpression of (4,6)-disulfated chondroitins and heparan sulfate (SKOV3 cell line) affects HP-evoked currents with En2 and Otx2 at 10 μM (Fig. 4C and D), a concentration for which no significant current could be detected in CHO-K1. It was reported that Otx2 targets disulfated chondroitin sulfate in perineural nets of the central nervous system (45). We used thus a cell line, SKOV3, enriched in this type of GAGs. In SKOV3 cells, we found that both En2 and Otx2 at 10 μM induced detectable currents (Fig. 4C and D and SI Appendix, Table S3). The total charge transfer induced by Otx2 was increased at all membrane potentials compared to those induced in CHO-K1 cells with supralinear effect at -55 and -75 mV (SI Appendix, Table S3); these effects were less prominent for En2. In addition, we measured a marked increase of Otx2-evoked currents for all hyperpolarized membrane states, as observed previously for CPPs. These results are in agreement with a recent report showing that a chondroitin sulfate-E targeting peptide, but not a heparan sulfate-targeting one, displays enhanced translocation in SKOV3 cells (43).

Our data show that HPs, En2 and Otx2, display translocation phenotypes similar to their related peptides but at higher concentrations (SI Appendix, Table S3), an observation in line with the fact that the cell-penetration of these HPs is highly regulated by the recognition of specific glycosaminoglycan cell-surface composition (45, 46).

A Toxic Cargo Peptide Conjugated to R_6W_3 Triggers Cytotoxicity After Translocation Process. A major biotechnological application of CPPs is their ability to transport various cargo molecules into cells (1, 8). Here, we explore whether a R_6W_3 -transported cargo would impact the translocation properties at play for the CPP alone and conjugated a reported cytotoxic peptide (KRAKLAK) (47) to R_6W_3 .

First, using a cytotoxic assay in CHO-K1 cells (Materials and Methods), we examined whether the KRAKLAK alone or conjugated to R_6W_3 was cytotoxic at various concentrations (5, 10, 20, and 50 μM) (SI Appendix, Fig. S5A). KRAKLAK alone, up to 50 μM , did not induce any cytotoxic effect with a cell viability unaffected, whereas its R_6W_3 -conjugated analogue (R_6W_3 -KRAKLAK) was cytotoxic from a concentration of at least 10 to 20 μM after overnight incubation with CHO-K1 cells (SI Appendix, Fig. S5A). In line with this, we observed that few minutes (from 3 to 15 min) of contact between the cell-attached pipette containing R_6W_3 -KRAKLAK and the cell membrane, were sufficient to promote cell death as illustrated by the loss of electrochemical gradient integrity (SI Appendix, Fig. S5B). Those effects result from the specific translocation of the transported cargo into the cytosol since KRAKLAK alone did not exhibit noticeable effect on membrane potential which remained stable around -25 mV (SI Appendix, Fig. S5B).

We next monitored the entry modalities of KRAKLAK alone vs R_6W_3 -KRAKLAK in CHO-K1 at a single-cell level (SI Appendix, Fig. S5C–F). At a $E_{\text{holding}} = -25$ mV, we observed that R_6W_3 -KRAKLAK induced transient currents at 10 μM but not at 1 μM ; note that at 10 μM R_6W_3 -KRAKLAK induces 4-times less total charge transfer compared to R_6W_3 , most likely reflecting the cargo hindrance (SI Appendix, Fig. S5C and D). KRAKLAK alone did not show significant total charge transfer when compared to the null peptide concentration condition (SI Appendix, Fig. S5C and D) indicating an absence of internalization of KRAKLAK alone. Regarding the voltage-dependence of R_6W_3 -KRAKLAK, we observed a marked asymmetrical relationship similar to R_6W_3 , for a 10 μM concentration, but not at 1 μM , for which no significant internalization could be detected (SI Appendix, Fig. S5E). Interestingly, at -75 mV the total charge transfer evoked by R_6W_3 -KRAKLAK was 4,000-times higher than at the RMP ($E_{\text{holding}} = -25$ mV) and also 4-times higher than total charge transfer induced by R_6W_3 at the same concentration (Fig. 5E and SI Appendix, Table S4). p_{off} estimation shows a rather stable average opening duration value remaining in the range of 0.7 to 1.2 ms for R_6W_3 -KRAKLAK-induced currents across the different E_{holding} (SI Appendix, Fig. S5F) similar to those observed for R_6W_3 and other CPPs and HPs (0.6 to 0.8 ms). As for the CPPs or HPs, the measured charge transfer ratio is supralinearly higher than the theoretical charge transfer ratio for each E_{holding} , indicating again that R_6W_3 -KRAKLAK induces the formation of additional transient pores (SI Appendix, Table S4).

Altogether, these results demonstrate that the R_6W_3 -KRAKLAK peptide translocates via transient pores into the cytosol to exert its cytotoxic effects. The R_6W_3 -cargo, when compared to R_6W_3 alone, shows a marked shift in terms of concentration ability to induce translocation, since a 1000-times concentration is required to observe detectable induced currents. However, it is worth noting, that once this concentration was reached the magnitude of the evoked-currents was 4-times greater for R_6W_3 -cargo than for R_6W_3 alone.

Homeoprotein-Induced Transient Currents in Pyramidal Cells of the Rodent Somatosensory Cortex. Last, we investigated whether the mechanisms of HP internalization are similar in a more physiological context, such as in acute brain slice preparation from adult mice (P50–60). We performed double patch-clamp (whole cell and cell-attached) in single pyramidal cell of the somatosensory cortex, in presence of the HP En2 in the cell-attached pipette at 1 or 10 μM (Fig. 5A). Pyramidal cells displayed a linear I/V (Fig. 5B).

Using identical extracellular and intracellular solutions as for ovarian cell lines, we first monitored the occurrence of En2-induced currents at RMP (RMP = -65 ± 6.9 mV, $n = 6$ pyramidal cells), then at hyperpolarized membrane potentials ($E_{\text{holding}} = -80$ and

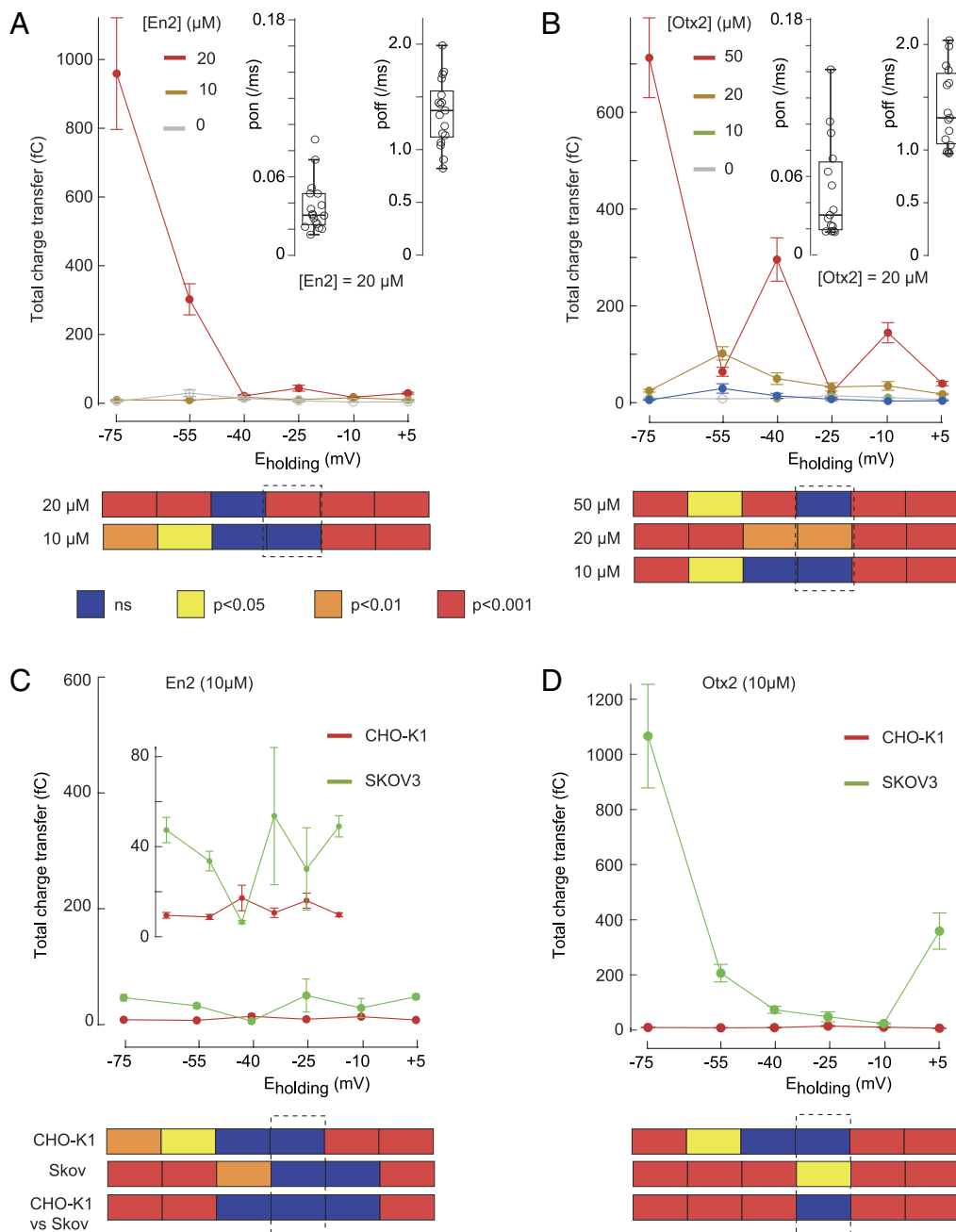


Fig. 4. HPs, Otx2 and En2, evoked transient currents favored by hyperpolarization and specific cell-surface glycosaminoglycans. (A and B) Total charge transfer measured at various E_{holding} show that HPs (A) En2 [10 μM: $n = 6$ cells; 20 μM: $n = 6$; $p_{\text{ANOVA}} = 2.56 \times 10^{-15}$ (concentration effect), 3.59×10^{-18} (E_{holding} effect)] and (B) Otx2 [10 μM: $n = 6$; 20 μM: $n = 6$; 50 μM: $n = 5$; $p_{\text{ANOVA}} = 7.44 \times 10^{-48}$ (concentration effect), 9.52×10^{-27} (E_{holding} effect)] induced transient currents from 20 μM concentration, which were favored by hyperpolarized membrane potentials. The null concentration data were obtained without CPP in the cell-attached pipette solution at each tested E_{holding} in CHO-K1 cells ($n = 5$). Insets: p_{on} and p_{off} were extracted (Materials and Methods) for En2 and Otx2 at $E_{\text{holding}} = -25$ mV, and both at 20 μM, a concentration for which unitary events could be distinguished; p_{off} show unitary events displaying submillisecond duration with a 12 to 16 Hz frequency range (p_{on}). One-way ANOVA show no significant difference between En2 and Otx2 for p_{on} ($P = 0.2629$) and p_{off} ($P = 0.6893$). (C and D) Total charge transfer related to E_{holding} in CHO-K1 and SKOV3 [overexpressing (4,6)-disulfated chondroitin and heparan sulfates] for 10 μM En2 [$n = 6$ and 4, respectively; $p_{\text{ANOVA}} = 5.89 \times 10^{-1}$ (E_{holding} effect), 9.42×10^{-9} (cell line effect)] and 10 μM Otx2 [$n = 6$ and 6, respectively; $p_{\text{ANOVA}} = 1.57 \times 10^{-17}$ (E_{holding} effect), 8.22×10^{-9} (cell line effect)] show increased transient currents mainly for hyperpolarized membrane potentials. Color-coded statistics above each graph indicate Student's t test values comparing the null concentration with the indicated CPP concentration (bilateral unpaired test, using Satterthwaite's approximation if unequal variances from F -test).

−95 mV (Fig. 5 C and D); it was not possible to apply significant depolarization since pyramidal cells RMP is close to their rheobase (Fig. 5B). At RMP, En2-induced transient currents in the same range (total charge transfer ≈ 100 fC) at both 1 and 10 μM. At 10 μM En2, there was a marked increase in total charge transfer at $E_{\text{holding}} = -80$ mV followed by a potent 38-times increase of the total charge transfer when pyramidal cells were hyperpolarized from −65 mV to −95 mV (Fig. 5 C and D); note that a limited hyperpolarization of −15 mV (from −65 to −80 mV) was sufficient to lead to a 5-times increase in the total charge transfer for both 1 and 10 μM En2. At −95 mV, the total charge transfer was also 20-times higher at 10 μM than at 1 μM. At 1 μM En2, we observed an increase of the total charge transfer at $E_{\text{holding}} = -80$ mV, which was not maintained when shifting to −95 mV (but still higher than at −65 mV).

Interestingly, the induced current frequency follows a similar trend as for CPPs, which is about 33 Hz at 1 μM and 200 Hz at

10 μM (Fig. 5E). The duration ($1/p_{\text{off}}$) of the En2-induced transient currents was submillisecond in line with those recorded for CPPs in cell lines, that is ~ 600 μsec at 1 μM (Fig. 5E); note that the opening duration was significantly shorter when compared to En2 at 20 μM in CHO-K1 ($E_{\text{holding}} = -25$ mV) (one-way ANOVA: $P = 0.0001$). p_{off} values exhibit no, or weak, dependency to the membrane potential, indicating that the observed events might be unitary translocation events. This observation indicates that the total charge transfer enhancement might result from an increase in the number of translocation events but not of the magnitude of the flowing current per translocation event.

Altogether, our data show the existence of unitary translocation events for En2 activity in neurons in acute brain tissues, which exhibits concentration and voltage dependencies similar to those of CPPs. This shows the robustness and conservation of the entry process, of CPPs and HPs, from cell lines in culture to neurons in ex vivo rodent brain slices.

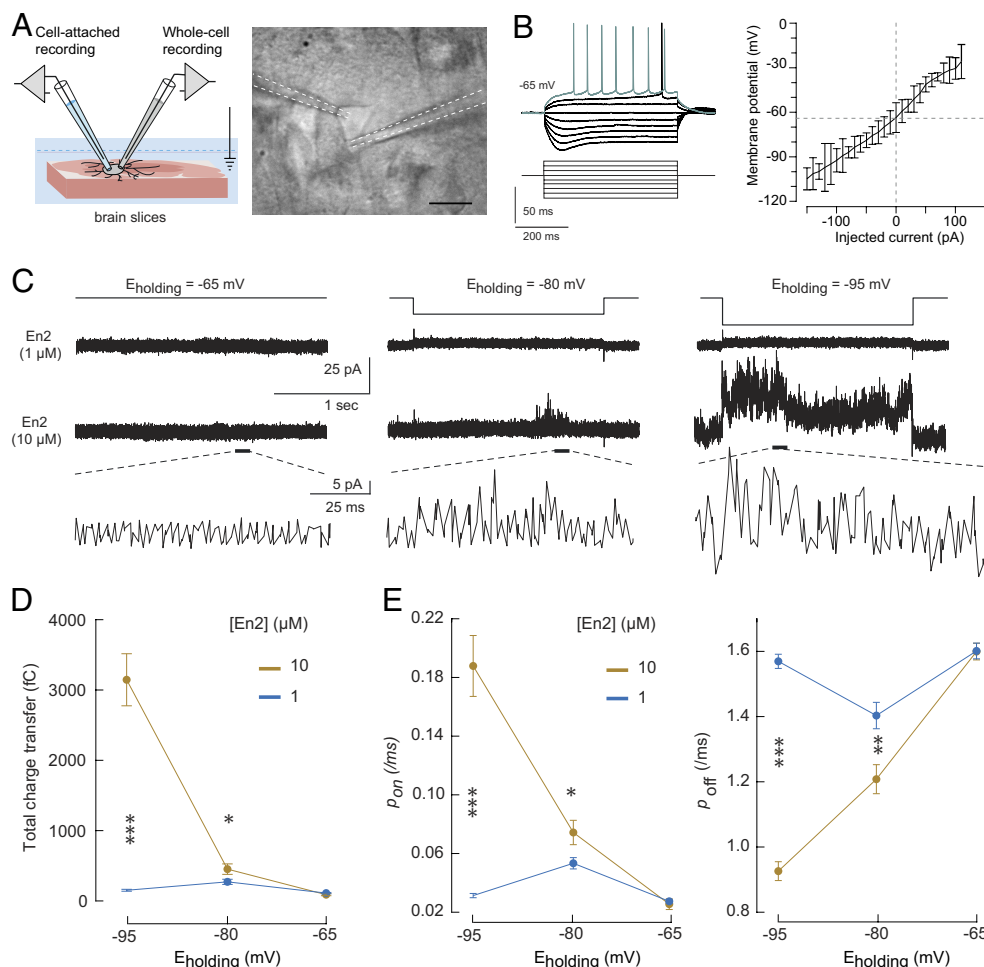


Fig. 5. Homeoprotein En2-induced transient currents in cortical pyramidal cells in mice. (A) Illustration and infrared microphotographs of double patch-clamp recordings of a pyramidal cell of the somatosensory cortex in mouse brain slice; pipettes are indicated in dashed white line. (Scale bar, 20 μm .) (B) Evoked-responses of a pyramidal neuron to current step injections (from -150 pA with 30 pA increasing steps) with the voltage traces in black up to rheobase and in blue for a spike train ($+30$ pA above rheobase). Pyramidal neurons display a linear I/V relationship with RMP ~ -65 mV ($n = 6$). (C) Representative raw current traces recorded in cell-attached mode with En2 at 1 and 10 μM , at different E_{holding} (-65 , -80 , and -95 mV), illustrate the increasing number and magnitude of En2-induced currents with increasing concentrations and hyperpolarization magnitude. (D) Total charge transfer depending on E_{holding} for En2 at 1 μM ($n = 3$) and 10 μM ($n = 3$). A massive increase of the total charge transfer was measured upon hyperpolarization, 38-times between -65 and -95 mV, for En2 10 μM ; note that a 5-times increase occurred for a limited hyperpolarization of 15 mV (from -65 to -80 mV) for both 1 and 10 μM En2. (E) p_{on} and p_{off} related to E_{holding} . At 1 μM , En2-induced transient currents have a duration of ~ 600 μs (similar to CPP in CHO-K1, see e.g., Figs. 1 and 4). Student's t tests were performed to compare the means of each couple of conditions.

Discussion

We explored the capacity of CPPs and HPs to translocate into cells through membrane perturbation events, i.e., transient pores. We hypothesized that in contrast to the general view of antimicrobial peptides (AMPs) mechanisms of membrane permeation, recently questioned (48), CPPs do not form stable structurally defined pores. Kinetics of the pore formation of AMPs in model lipid systems are within tens of seconds (49), far out of the kinetical range observed herein. We cannot discard that vesicle budding and collapse (50) might occur; such events, detected by time-lapse confocal imaging, arose however about 1 min following the addition of CPPs.

We developed a single-cell double patch-clamp protocol that permits simultaneously to control the membrane potential of one cell, while monitoring the ion fluxes induced by local application of CPPs or HPs to a restricted membrane surface area ($1 \mu\text{m}^2$). This specific set-up allowed to monitor translocation-induced transient currents evoked by CPP or HP. We established that the CPPs Tat, R_9 , penetratin, R_6W_3 , and the HPs Otx2 and En2 induce transient unitary currents with kinetics that were similar in cell lines (RMP: -25 mV) and pyramidal cells of brain slices from adult mice (RMP: -65 mV). The total charge transfer reflects the passage of the peptides and proteins across the plasma membrane as we evidenced by the fluorescence imaging of CF- R_6W_3 and the R_6W_3 delivery of a toxic KRAKLAK cargo molecule inside cells. In all cases, imposing 15 mV hyperpolarization steps to the cells strongly increased the CPP- or HP-evoked total charge transfer. Therefore, limited hyperpolarization such as 15 mV can considerably boost by 10 to 100 times the CPP/HP translocation.

The cell-attached configuration allows the detection of events in the submicromolar CPP range. We used a high sampling rate (20 kHz), which permits the recording of cellular currents occurring within a 50 μs time range. This provided us with high temporal resolution of ionic fluxes, and enables to detect and characterize unitary ionic events as brief as the typical opening durations of voltage-dependent ionic channels (from one to hundreds of milliseconds) (51), or “kiss-and-run” fusion of vesicle at synapses (from 10 milliseconds to 1 s) (52). Cell-attached patch-clamp recordings in voltage-clamp mode have been initially used for revealing and characterizing single ionic channel activity at a submillisecond time scale (53). The duration of our recorded unitary events was between 500 to 800 μs , which is shorter than the voltage-induced single pores detected in lipid bilayer, with 3 ms for the shortest opening (54). In addition, these pores were observed under a high electric field from 200 to 500 mV (54), which is 5- to 10-times higher than the polarization we applied. Importantly, even without polarization, we observed transient pores induced by CPPs and HPs in CHO-cell lines or acute rodent brain slices. Overall, the transient currents observed here upon CPP or HP application display a submillisecond kinetics scale, that is below the duration of all reported fast/ultrafast endocytosis events at central fast calyceal synapses (100-times shorter): the time constant reported after a single vesicle fusion is 56 ms at low neuronal spiking activity (<2 Hz) and considerably increases up to tens of seconds for higher firing rate (55). More generally, duration of the fast endocytosis open phase (also called the Ω -shaped membrane phase) is around few hundreds of milliseconds with membrane capacitance measurement (56–58). Slow

endocytosis is even out of the time scale of the observed submillisecond transient pores since they last over from 1 to 30 s, this duration corresponding to the vesicle open phase (56, 59, 60).

Depending on the CPP and the imposed membrane potential, the total charge transfer covers a wide range of values, from a few dozen to several thousand fC. If one considers that the total charge transfer might correspond to the strict transport of positively charged peptides only, this would imply an intracellular concentration of 1 to 10 mM peptides, which is most unlikely. We previously evidenced that incubation of cells for one hour with 1 to 10 μM extracellular concentrations of CPPs led to intracellular levels of peptides in the μM range (3, 10, 32, 61), which is about 100- to 1,000-times less than the total charge transfer measured herein for any of the CPPs. We can thus conclude that the total charge transfer results from ion fluxes through the transient pores formed within the cell membrane by the CPPs and HPs (SI Appendix).

The magnitude of total charge transfer evoked by the CPPs or HPs is not related to their net charge, since the efficiency range of each peptide to evoke total charge transfer varies according to their concentration and the imposed membrane potential: for example, at RMP and 10 μM [CPP], the total charge transfer value range was $R_9 \approx R_6W_3 > \text{Tat} > \text{penetratin}$ (net charge $9 \approx 6 > 8 > 7$); at -75 mV and 1 μM [CPP], it became $R_9 > \text{penetratin} \approx R_6W_3 > \text{Tat}$ (net charge $9 > 7 \approx 6 > 8$); at -75 mV and 10 μM [CPP], the total charge transfer ranking became $\text{Tat} > R_9 > R_6W_3 > \text{penetratin}$ (net charge $8 > 9 > 6 > 7$). These results are likely part of explanations for the different penetrating or delivery efficiency of CPPs reported in the literature according to the cell type (and consequently RMP) and experimental conditions (e.g., peptide concentration) used (62). In addition, we observed that the CPP or HP passage of the membrane occurs from the exterior of cells directly to the cytosol, that is by direct translocation. Our data supporting such assertion are the followings:

- i) Unitary pores formed by a CPP last for 500 to 800 μsec , a time range about 1,250- to 40,000-times lower than for the classical endocytic (vesicle open phase) processes (1 to 20 s) (38); the fact that we might have measured faster endocytosis processes such as macropinocytosis is also unlikely because the size of these membrane engulfment-formed vesicles is typically 350 to 1500 nm, which corresponds to a membrane surface of 0.12 to 28 μm^2 to be compared to the 1 μm^2 of the patch-clamped membrane and in any case the duration of these events are within the 5 to 10 s time range (40); even clathrin-independent ultrafast endocytosis observed at neuronal synapses have also time duration about 100-times higher (50 to 100 ms) (20, 63, 64).
- ii) Formation of the pores induced by CPPs or HPs occur every 500 ms within the 1 μm^2 membrane, when compared to endocytic processes observed in fibroblasts or epithelial cells, in which one new clathrin-coated pit per μm^2 is formed every 2 min (20). We estimated the size of the transient pores induced by CPPs or HPs, and assessed a 0.5 nm radius, taking the current $I = g \Delta V$ and the conductivity $g = (\pi * d / 2 * s) / (4 * e)$, with $e = 5$ nm pore depth, $s = 1.4 \text{ S.m}^{-1}$ conductivity in 150 mM NaCl, $I = 10$ pA and $\Delta V = 50$ mV. This radius value is identical to the one reported for R_9 -evoked transient pores in model membrane of 20/80 DOPS/DOPE (65). Importantly, pore radii induced by CPPs are two- to three-fold smaller than those formed by AMPs, contributing to their instability due to the strong elastic restoring forces that result from the high local membrane curvature (65).
- iii) When all endocytic processes are inhibited at 11 $^\circ\text{C}$, transient pores still formed during the passage of CPP, with similar kinetic properties as those we recorded at 34 $^\circ\text{C}$; decreasing the

temperature affects only the frequency of pore formation, but not their opening duration. In parallel, fluorescence imaging brings evidence for cell penetration at 22 $^\circ\text{C}$, a temperature at which endocytosis processes are slowed down, up to about 10-times compared to the endocytosis rate at 35 $^\circ\text{C}$ (66).

- iv) The nonpenetrating cytotoxic peptide, KRAKLAK, is efficiently delivered by R_6W_3 with the same trend of concentration, voltage-dependence, and transient current kinetics than the CPP alone, and leads to cytotoxic effects.
- v) Endocytosis processes are unaffected by membrane potential hyperpolarization or depolarization shifts (29, 37).

The evaluation of the impact of the electrochemical gradient in the magnitude of the total charge transfer indicates that the polarization and concentration of CPPs/HPs induced a supralinear effect, highlighting the recruitment of an extrapool of CPPs/HPs for the formation of additional pores.

We also observed that depolarization might occasionally give transient evoked-events. The fact that R_6L_3 also induced such transient events during depolarization (but not hyperpolarization) (SI Appendix, Table S1) could account for an increased background noise reflecting nonphysiological resting membrane potential values.

Of utmost interest, we found that the presence of cell-surface GAGs is mandatory for the translocation of CPPs and HPs, a direct role that has been rarely reported in previous studies (15, 43, 46, 67). Cell-surface GAGs play several crucial roles in cells, as binding sites or receptors for a variety of ligands, such as growth factors or chemokines, facilitating their internalization through endocytosis or cell signaling events. For HPs, Otx2 and En2 internalization in specific brain regions is dependent on the presence of specific GAGs in the extracellular matrix (43, 46, 68). Otx2 interacts preferentially with (4,6)-disulfated chondroitin sulfates in perineural nets to internalize in cortical parvalbumin interneurons (45). En2 interacts preferentially with heparan sulfates as a first step of internalization (46). A glycosaminoglycan-binding sequence was identified in Otx2 and En2. Although positively charged, the respective glycosaminoglycan binding sequence of Otx2 (RKQRRETTTFTRAQL) and En2 (RSRKPKKKNPKNKEDKRPR) does not penetrate into cells per se but confers additional cell entry specificity to a conjugated penetratin sequence (43).

Our observation about the key role of GAGs in translocation helps to solve the conundrum regarding the CPP translocation mechanism. Indeed, studies used generally lipid bilayer models (thus deprived of GAGs), which necessitate nonphysiological conditions to promote CPP translocation. CPPs can adsorb at the surface of lipid bilayers through electrostatic interactions and hydrophobic interactions. The insertion of CPPs into the bilayer depends on a balance between enthalpic gains from favorable hydrophobic interactions and entropic costs from the loss of conformational freedom. In purely synthetic lipid models, a significant proportion of anionic lipids (e.g., phosphatidylglycerol or phosphatidylserine) or lipids with negative intrinsic curvatures (e.g., phosphatidylethanolamine or ceramide) are generally used. With these models, positively charged amino acids (e.g., Arg) interact with negatively charged phospholipids, allowing hydrophobic amino acids (e.g., Trp) to intercalate within lipid acyl chains. Lipids having spontaneous negative curvature (14, 65, 69, 70) lower the energy barrier for CPPs to intensify the curvature and disrupt the bilayer to create a transient pore. In such models, addition of cholesterol, a major component of eukaryotic cell membrane, has crucial effects by reducing the packing of unsaturated and saturated phospholipids (71), thus destabilizing the capacity of CPPs to induce a cubic phase from the phospholipids

with negative curvature. Without negatively charged phospholipids or phospholipids with negative intrinsic curvatures, insertion, and bilayer perturbation by positively charged CPPs are hardly detectable (65, 72, 73), although CPPs still adsorb onto the lipid bilayer. However, with zwitterionic phospholipids (e.g., phosphatidylcholine or sphingomyelin) only (25, 73, 74), positively charged amino-acids in CPPs can interact directly with the phosphate group of the polar head of these phospholipids. This latter interaction results in shielding CPPs positive charges and in more hydrophobic CPPs able to transfer at lower energy costs into the hydrophobic core of the lipid bilayer through a cubic phase, a process akin to counterion-assisted phase transfer of CPPs.

In this context, as we evidenced herein, GAGs appear as the missing piece of the mosaic translocation process through transient pore formation (*SI Appendix*, Fig. S6). We showed that chondroitin sulfates interact indeed with the choline headgroups of the lipid bilayer (43), the bilayer being stabilized by Ca^{2+} bridges between phosphate groups. The negatively charged linear polymer acts as a gateway between the positively charged CPPs and the phosphate groups of the lipid bilayer, allowing the concentration of CPPs at the vicinity of the hydrophobic core of the membrane. The electrostatic interactions between positively charged heads of phospholipids (choline) and negatively charged carboxylate and sulfate groups of GAGs, reduce the energy barrier for peptide binding and insertion into the bilayer and thus facilitates the translocation of CPPs through transient pores (radius of ≈ 0.5 nm), by exploiting membrane defects or the boundaries between disordered and ordered membrane domains. Since the cell plasma membrane contains thousands of lipid species, which endow the membrane with tension, bending rigidity and dynamic phase properties, the lipid identity within the pore or in its immediate surroundings needs further investigation. From the inner leaflet side, PIP2 is an efficient trap for positively charged peptides (75) and a key partner for translocation (76). Since PIP2 also anchors and regulates the cortical actin network and controls membrane tension and curvature (77), we assume that pore formation at the boundaries of these cytoskeleton-anchored dynamical membrane nanodomains, is also energetically favored.

Finally, a hypothesis issued from a modeling investigation based on cell membrane lipid content, (but without GAGs), posits that CPPs would induce a megapolarization of ~ 2 V accounting for the CPP crossing of the lipid bilayer (29). To our view, this putative megapolarization would be similar to electroporation leading to nanoseconds opening of pores. In the present study, we solve this question by showing that the presence of GAGs is mandatory for CPP and HP translocation, and their absence at the cell-surface cannot be compensated by a strong membrane polarization. In addition, we observed that translocation takes place at holding membrane potentials close to RMP, an experimental condition that prevents the occurrence of any megapolarization step.

The translocation efficacy depends on the CPP/HP sequence, the membrane potential, peptide concentration, and GAG composition. By taking into account this whole range of parameters

one could improve both cell-selectivity and cell delivery efficacy of CPPs/HPs and sustainably impact their use for biotechnological and therapeutical applications. Finally, since we show that a CPP–cargo conjugate has similar translocation characteristics as the CPP alone or as other CPPs and HPs and that this translocation process happens similarly in different cultured cell lines and in more physiologically relevant brain slice model, we might foresee new targeted delivery strategies for biotechnological and therapeutical applications. Since many therapeutic targets are cytosolic and cell-type specific, the exact knowledge of the biophysical and biochemical features of these cells would help optimizing the design and use of specific CPPs/HPs (and bioactive conjugates) as the optimal candidates for therapeutic drug delivery.

Materials and Methods

Detailed *Materials and Methods*, that is, patch-clamp and two-photon recordings, cytotoxicity assays, and data analyses are included in the *SI Appendix*.

Experiments were conducted in CHO-K1, CHO-pgsA-745, and SKOV3 cells or acute brains slices from adult mice.

Simultaneous Whole-Cell and Cell-Attached Recordings. Single CHO Cell or Pyramidal Cell Was Recorded in Cell-attached (pipette Containing CPP or HP) for the detection of transient CPP/HP-evoked transient currents and whole-cell to impose voltage steps. Signals were amplified using EPC10-2 amplifiers (HEKA Elektronik) and sampled at 20 kHz with the Patchmaster v2 X 32 software (HEKA Elektronik).

Two-Photon Imaging. Data were acquired using the TriM Scope Matrix microscope (LaVision Biotech), combined with cell-attached patch-clamp (pipette containing CF-CPP or CF alone).

Transient Current Analyses. The raw current traces were first segmented as successions of open and close states allowing to compute their duration, amplitude, and unitary charge transfer. p_{on} and p_{off} were extracted using Bayesian inference of the current event dynamics.

Data, Materials, and Software Availability. All data are included in the article and/or *SI Appendix* and/or Zenodo repository (the data consists in patch-clamp recordings of the different cell-types and experimental conditions used) (78).

ACKNOWLEDGMENTS. This research has received fundings from the Agence Nationale de la Recherche (Grants No. ANR-17-CE11-0050-CROSS and ANR-20-CE44-0018-GLYCOTARGET).

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The authors declare no competing interest.

This article is a PNAS Direct Submission. M.O. is a guest editor invited by the Editorial Board.

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